# LD + Frozen EmbeddingGemma Walk-Text Embeddings (WalkLM-Style) + SAGN/SLE + C&S on `ogbn-products`: A Validation-Locked Ten-Seed Study

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**Abstract**

We present an external-data node-classification pipeline for Open Graph Benchmark `ogbn-products`. It concatenates released Label Deconvolution BERT features, frozen EmbeddingGemma representations of graph-walk text, and raw OGB features, then trains a four-stage SAGN/SLE predictor and applies Correct & Smooth. A protocol-locked v2 run over ten fresh seeds obtains validation accuracy $0.940584 \pm 0.000433$ and test accuracy $0.874984 \pm 0.000490$ using unbiased sample standard deviation. The v2 validation grid and selected lock were committed before one guarded final-test invocation. We disclose that earlier v1 test-aware exploration informed the frozen encoder and SAGN configuration.

## 1 Dataset and evaluation

We use the official `ogbn-products` split and OGB evaluator from OGB 1.3.6. After SAGN preprocessing, the graph has 2,449,029 nodes, 123,718,280 edges, 196,615 training nodes, 39,323 validation nodes, 2,213,091 test nodes, and 47 classes. Ten v2 runs use seeds 10 through 19. Seeds 0 through 9 had already been consumed during test-aware v1 exploration, so v2 preregistered a fresh range and reports every run without selection. Means and unbiased sample standard deviations use the $n - 1$ denominator.

## 2 Method

The full method name is *LD + frozen EmbeddingGemma walk-text embeddings (WalkLM-style) + SAGN/SLE + C&S*. External data is declared as Yes.

### 2.1 Input representation

| Component | Dimension | Origin |
|---|---|---|
| LD BERT hidden state | 768 | Released `products_sagn/hidden_state.pt` from the LD feature release |
| Frozen EmbeddingGemma walk-text | 768 | Mean-pooled, normalized graph-walk text embeddings |
| Raw OGB feature | 100 | Original `ogbn-products` node feature |
| Concatenated input | 1,636 | Order: LD, walk-text, raw |

### 2.2 Walk-text extraction

For every node, four directed random walks of length four are sampled with seed 42. The PyG/OGB loader represents 61,859,140 undirected product relations as 123,718,280 directional edge records; the walks use that as-loaded ordering. Each visited product contributes the first 160 characters from extttnode-text.csv.gz, the GLEM title-and-description text distributed by the PyG extttTAGDataset path. Nodes are separated by "cites." The entire rendered text is tokenized to at most 160 tokens. Consequently, the first walk fills the input and walks two through four do not reach the encoder.

The final hidden state of `google/embeddinggemma-300m` is attention-mask mean pooled and L2 normalized. EmbeddingGemma is loaded with `AutoModel` and is not fine-tuned, unlike the language-model fine-tuning in WalkLM. "WalkLM-style" describes only the graph-walk textualization. The reconstruction pins model revision `57c266a7`; the original resolved revision was not captured.

The original modified generation source was not committed. The recovered command specifies `-mlm_epochs 0` and `-epochs 1`; the former keeps the encoder frozen, while the latter trains downstream diagnostic classifiers for one epoch and does not affect the saved embeddings. The script now provided is explicitly a post-hoc reconstruction. No v2 walk-text-only ablation was run, so the complete-pipeline difference cannot be attributed to this component alone.

### 2.3 SAGN/SLE and C&S

The predictor uses two feature-propagation hops, nine label-propagation hops, 512 hidden channels, one attention head, learning rate 0.001, batch size 50,000, and dropout values 0.5/0.4/0.2/0.5. Four self-training stages run for 300, 100, 100, and 100 epochs with thresholds 0.95, 0.95, and 0.999. The validation-selected C&S configuration uses 50 correction and 50 smoothing iterations, DAD/DAD normalization, correction alpha 0.35, smoothing alpha 0.60, and scale 20.

Only training labels enter SAGN/SLE label features and C&S propagation. Validation labels are used for early stopping and model selection. Test labels are not passed into propagation.

## 3 Results

| Evaluation | Validation | Test |
|---|---|---|
| V1 C&S, exploratory | $0.940010 \pm 0.000299$ | $0.874753 \pm 0.000487$ |
| V2 stage 3 before C&S | $0.939204 \pm 0.000522$ | Not evaluated |
| V2 validation-selected C&S | $\mathbf{0.940584 \pm 0.000433}$ | $\mathbf{0.874984 \pm 0.000490}$ |

The v2 test mean is 0.001284 above the public `GLEM+GIANT+SAGN+SCR` entry of $0.8737 \pm 0.0006$ recorded on September 4, 2026. The other comparator discussed here is `LD+GIANT+SAGN+SCR`, reported at $0.8718 \pm 0.0004$. Official status and ranking are determined by OGB.

The winning C&S candidate exceeds the runner-up by 0.0000762912 mean validation accuracy, exactly three validation nodes per seed on average, or 30 additional correct predictions summed over ten seeds.

# 4 Parameter accounting

| Component | Parameters |
|---|---|
| LD `bert-base-uncased` encoder | 109,482,240 |
| LD SAGN/SLE head and aggregation weight | 1,154,656 |
| EmbeddingGemma encoder | 307,581,696 |
| Final 1,636-input SAGN/SLE predictor | 5,010,014 |
| Correct & Smooth | 0 |
| Total | 423,228,606 |

The first two rows sum to the published 110,636,896 parameters for `LD+GIANT+SAGN+SCR`. Every preserved training log reports the final SAGN runtime count.

# 5 Reproducibility

The repository pins SAGN and LD revisions, packages the feature-integration patch, records both external tensor hashes and sizes, and provides an explicit EmbeddingGemma reconstruction. The original EmbeddingGemma generation versions, internal compute dtype, GPU, and resolved model revision were not captured and are reported as unknown rather than inferred from later runs.

A 128-node deterministic prefix was reconstructed with the pinned model and settings on an NVIDIA RTX PRO 4500 Blackwell. All 98,304 float32 embedding values exactly match the corresponding archived tensor rows (mean cosine similarity 1.0, mean L2 difference 0.0, maximum absolute difference 0.0). The comparison JSON and reconstruction environment are committed in the repository.

A v2-from-release script downloads all ten predictions, verifies their SHA-256 values, recomputes the validation grid and lock, and reproduces the final metric separately. Small v1 provenance artifacts have been migrated into this repository; large v1 assets are mirrored in an archived release. The public v2 release contains the predictions, hashes, environment, and log bundle. Each seed log ends with an upstream SAGN NumPy summary traceback after the stage-3 checkpoint has already been saved; the wrapper verifies the saved checkpoint before continuing.

# 6 Protocol audit

The v1 experiment compared language encoders, feature stacks, SAGN settings, and C&S candidates while test accuracy was visible. The final SAGN values, including $K = 2$, hidden size 512, epoch schedule, and batch size, were carried forward from this test-aware work. A conservative lower bound is 300 v1 test evaluations from the final ten-seed SAGN training alone; additional encoder and C&S test looks were not fully preserved.

The 27-member v2 C&S grid is identical to the focused v1 grid. V1 selected $(0.30, 0.65, 25)$ using test results; that candidate ranks 11th of 27 on v2 validation. V2 selected $(0.35, 0.60, 20)$ using validation only.

The preregistration froze fresh seeds, the inherited model, the grid, and the selection rule before v2 execution. Training masked all test labels and reported no test metric. Every candidate record declares `test_metrics_accessed: false`. The records and selected lock were committed before the guarded final-test command, which wrote a durable marker before dataset access. No other v2 test evaluation is present in the preserved commands, logs, marker, or repository history.

This creates a forward test-isolated boundary but cannot erase v1 method selection. The complete controlling disclosure is `TUNING_AND_PROTOCOL.md`.

# 7 Conclusion

Across ten fresh seeds, the validation-locked pipeline obtains $0.874984 \pm 0.000490$ test accuracy. Its predictions, validation candidates, lock, test-access marker, and aggregate arithmetic are auditable. Any OGB submission must declare external data and include the v1 historical disclosure.

# References


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